

NeuroShield – AI Powered Blood Brain Barrier Permeability Predictor



Team



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Problem



Many drug candidates developed for neurological diseases **fail** during laboratory and clinical stages because they cannot cross the **Blood-Brain Barrier (BBB)**, leading to **significant loss of time** and cost.





Solution/Product



NeuroShield is an **AI-powered** web platform that **predicts** whether a drug candidate can **cross the Blood-Brain Barrier** and reach the brain **before costly laboratory testing**. By giving researchers **fast** predictions, **clear** explanations, and **optimization insights**, it helps them focus on the most promising neurological drug candidates.





Demo/How It Works



Let's See Our Demo Video!

https://drive.google.com/file/d/1qLbmk0b11IEcjAVd1O594oTvFf_C33Vb/view?usp=share_link



Value & Impact



NeuroShield reduces the **time, cost, and early failure risk** in neurological drug discovery by **predicting BBB permeability before laboratory testing.**

It enables researchers to make **faster, data-driven** decisions and **prioritize the most promising** brain-targeting compounds.



Business Model



NeuroShield operates as a **B2B** and **B2G SaaS** platform for **pharmaceutical** companies, **biotech** startups, and **research institutions** that need **faster** early-stage CNS drug screening. Through **subscription-based access**, **enterprise API integration**, and **institutional licensing**, the platform can scale **globally** as an **AI decision-support tool** in **neuropharmaceutical R&D**.



Next Steps



Future versions of **NeuroShield** will integrate **deeper molecular learning models, larger pharmacological databases, and multi-parameter drug safety screening beyond BBB prediction.**

We also plan to expand the platform into a **full AI-assisted neurodrug discovery environment with molecule optimization, batch screening, and collaborative research reporting tools.**



Final Cover



NeuroShield redefines the future of neurodrug discovery (one of neuroscience's **most expensive unknowns**) by replacing costly trial and error experimentation with **instant, explainable, AI-driven** BBB intelligence; empowering researchers to identify brain-viable compounds **faster, smarter, and with far greater scientific confidence.**